IMPERIAL

Carbon Accounting and Reporting

Lecture 7 – Net Zero
Course Homework 2

Dr James Robey / Jon Hampson

31st October 2025

In the news…

Course Overview

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COURSE OVERVIEW

I. Sustainability – the Big Picture
II. Carbon and Climate Change
III. Basics of Carbon Accounting
IV. Introduction to Organisational Carbon Accounting
V. Scope 2 – energy emissions
VI. Scope 3 – indirect emissions
VII. Net Zero, Science Based Targets and Carbon Credits
VIII. Sustainability reporting & materiality

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Net Zero

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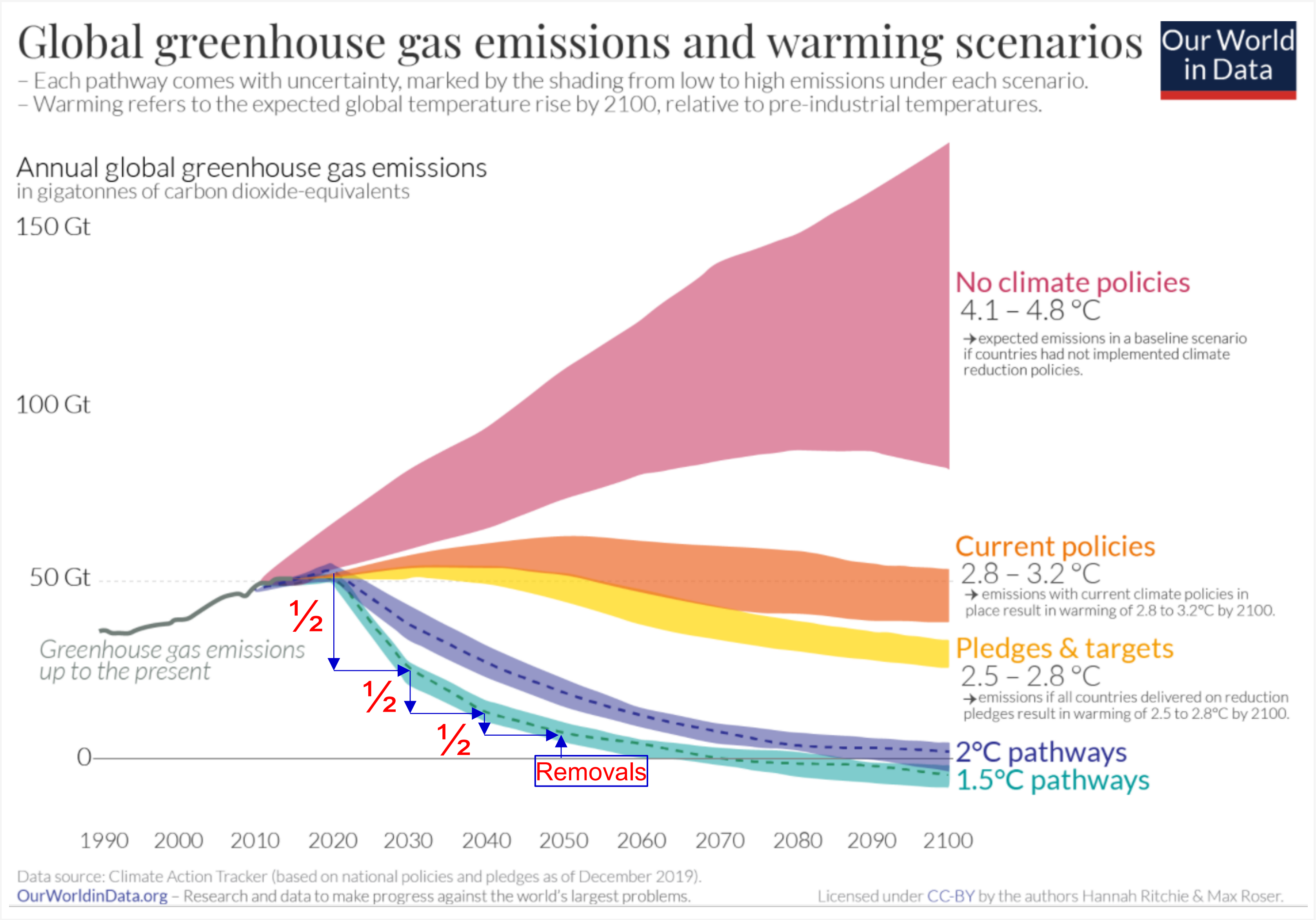

https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions

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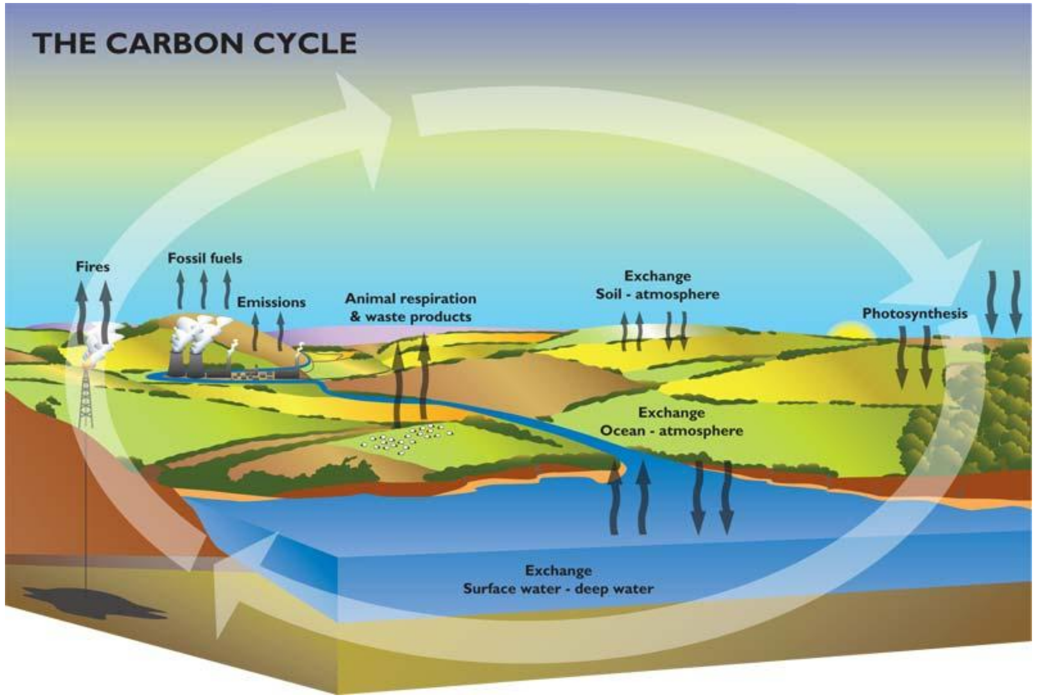
Net zero as a concept

There is still not one definition of net zero

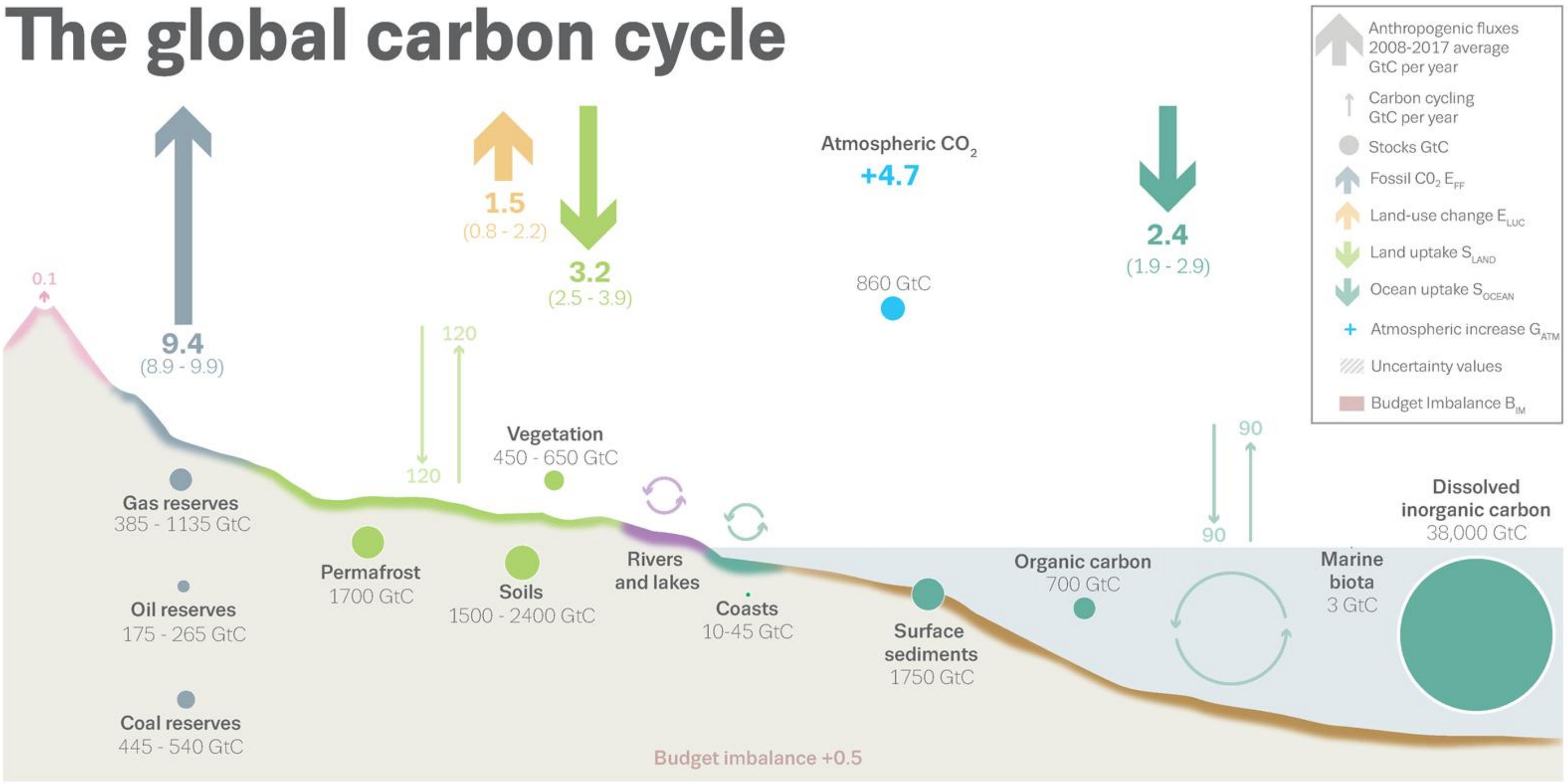
IPCC

“Net zero emissions are achieved when anthropogenic emissions of greenhouse gases to the atmosphere are balanced by anthropogenic removals over a specified period.”



Source: British Geological Survey

It is not all about fossil fuels



https://doi.org/10.18160/gcp-2018

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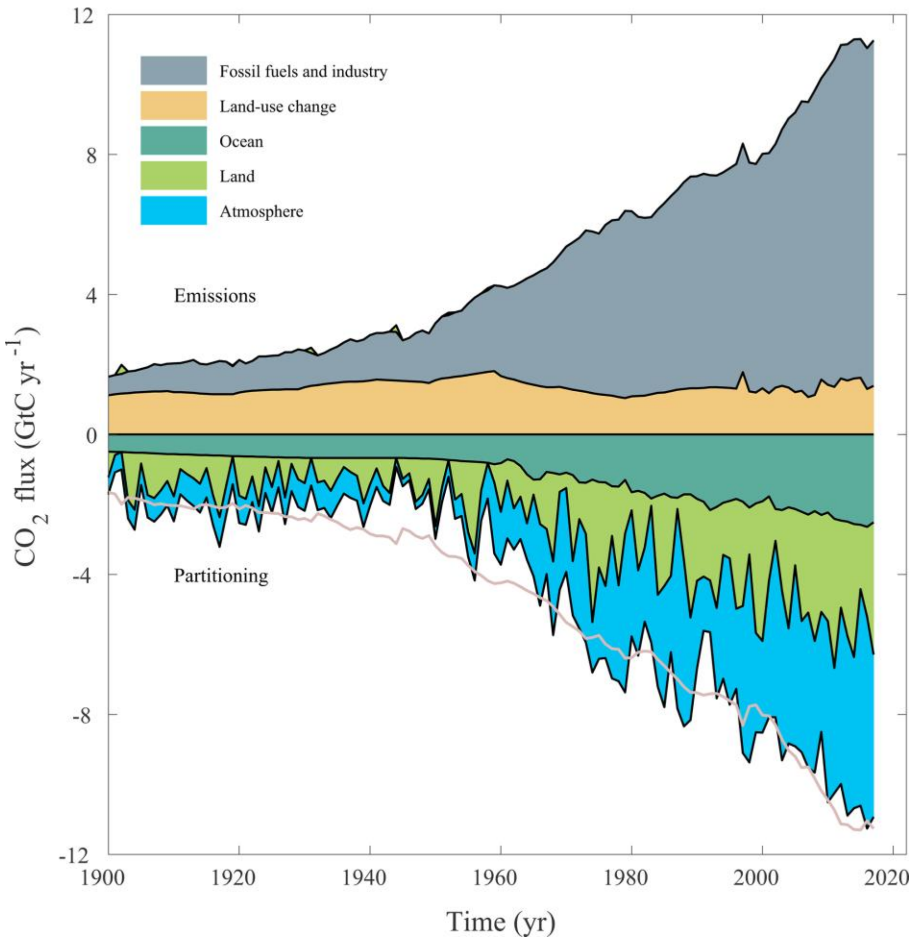
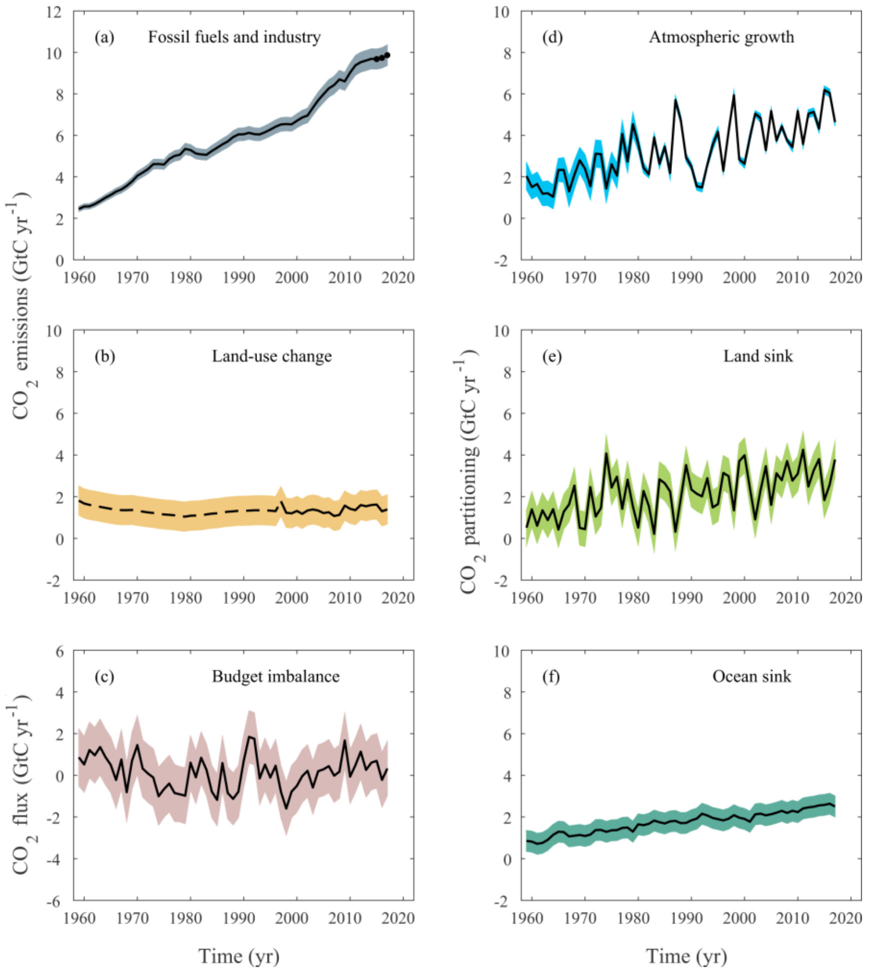
The balance of carbon





https://www.earth-syst-sci-data.net/10/2141/2018/

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Net zero as a concept

CDP and SBTi:

"Achieving a state in which the activities within the value chain of a company result in no net impact on the climate from greenhouse gas emissions. This is achieved by reducing value chain greenhouse gas emissions, in line with 1.5°C pathways, and by balancing the impact of any remaining greenhouse gas emissions with an appropriate amount of carbon removals."

SCIENCE BASED TARGETS
DRIVING AMBITIOUS CORPORATE CLIMATE ACTION

CDP
DISCLOSURE INSIGHT ACTION

CARBON TRUST

https://www.carbontrust.com/news-and-events/insights/net-zero-an-ambition-in-need-of-a-definition

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Net zero by when?

IPCC

“Limiting warming to 1.5°C implies reaching ***net zero CO_2 emissions globally around 2050*** *and concurrent deep reductions in emissions of non-CO_2 forcers, particularly methane”*

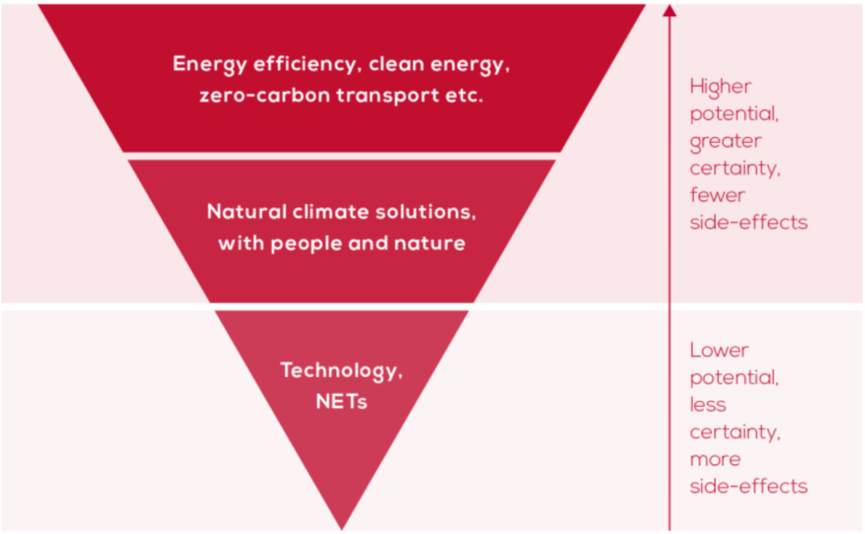
https://www.ipcc.ch/site/assets/uploads/sites/2/2019/02/SR15_Chapter2_Low_Res.pdf

Why "net" zero?

1. **Zero** is not likely to be possible across all sectors of the economy

2. **CO_2** is the only greenhouse gas that can be easily absorbed from the atmosphere

3. **Negative emissions** i.e. taking greenhouse gas emissions out of the atmosphere will be needed



https://eciu.net/analysis/briefings/net-zero/net-zero-why

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Net zero commonly discussed at three levels

Buildings

Companies

BNP PARIBAS - 2020

BOSCH - 2030

IKEA - 2030

Unilever - 2030

sky - 2030

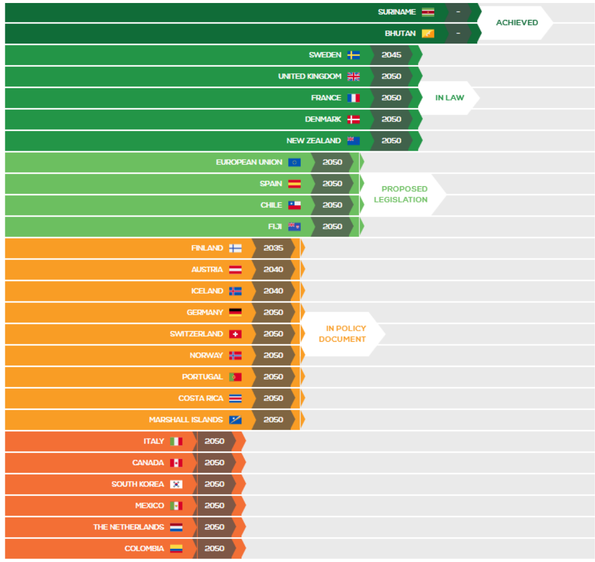
SAP - 2025

centrica - 2050

MAERSK - 2050

TATA STEEL - 2050

Countries & Cities



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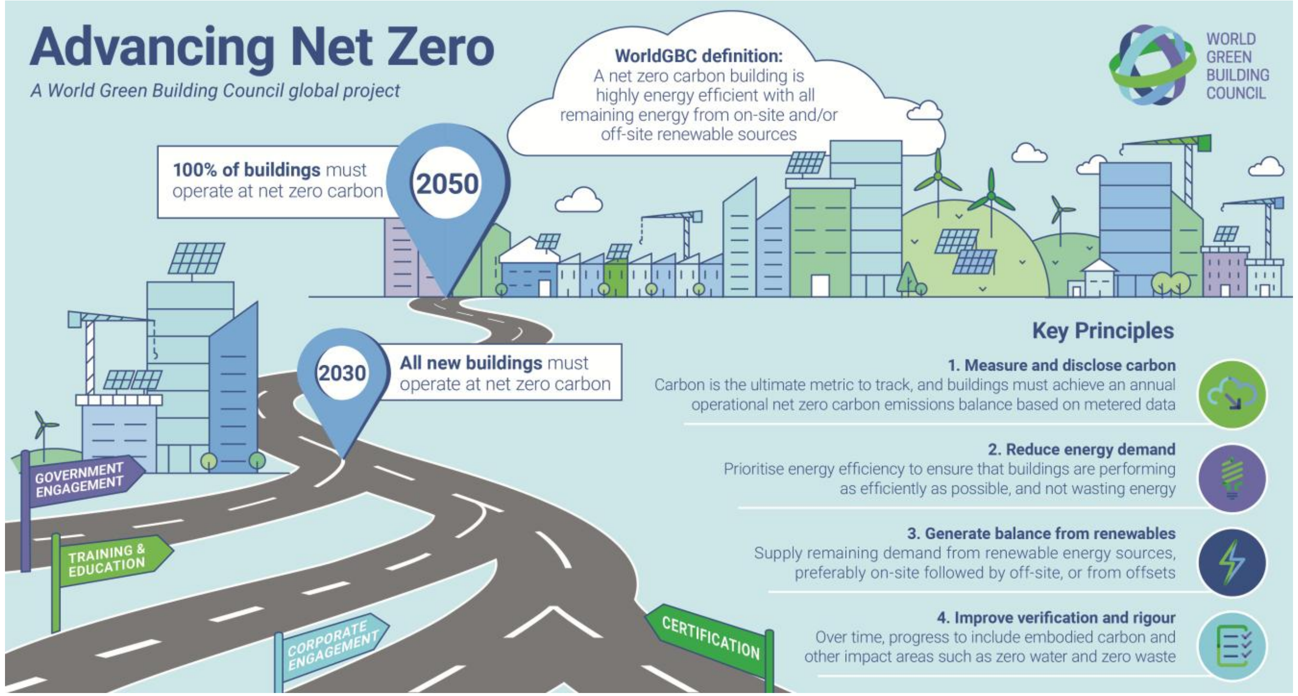
Net zero for buildings



1 Commit
Advanced trajectory for all new and existing buildings within direct control of the organisation to operate at net zero carbon by 2030

2 Disclose
Measure, disclose and assess annual asset and portfolio energy demand and carbon emissions

3 Act
Develop and implement a carbon roadmap outlining key actions and milestones

4 Verify
Demonstrate enhanced energy performance, reduced carbon emissions & progress towards net zero carbon assets and portfolio

5 Advocate
Demonstrate leadership to support the transition towards net zero carbon buildings

WGBC definition

Net zero carbon is when the amount of carbon dioxide emissions released on an annual basis is zero or negative. Our definition for a net zero carbon building is a highly energy efficient building that is fully powered from on-site and/or off-site renewable energy sources and offsets.

76 companies, 28 cities, 6 states or regions

salesforce
Goldman Sachs
HUDSON PACIFIC PROPERTIES
JLL
MULTIPLEX Built to outperform.

https://www.worldgbc.org/thecommitment

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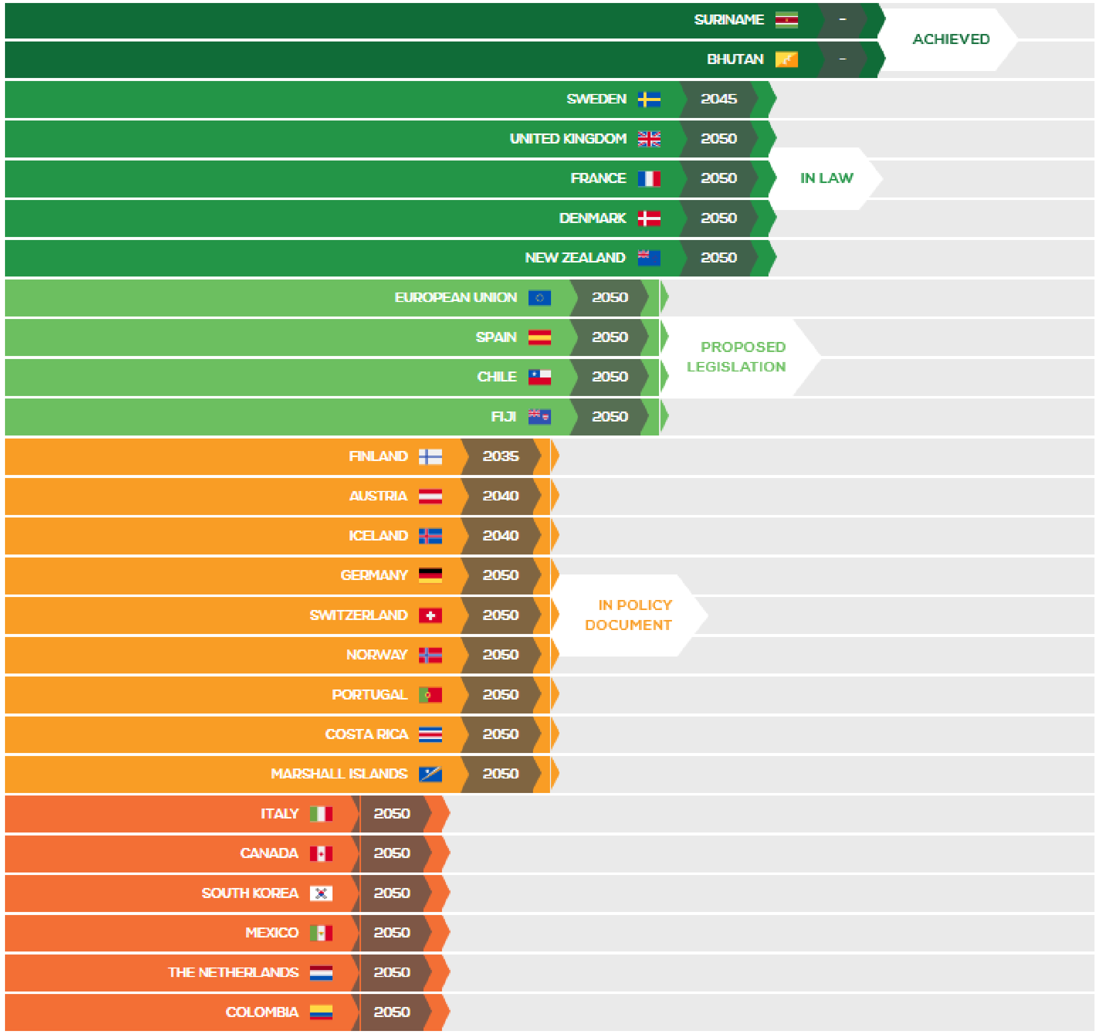
Net zero for countries



Key questions

1. What is being included? Does the target cover all GHG emissions and all sectors?

2. Are the reductions happening within the country?

3. How is accounting for anthropogenic removals being handled? For example, are all the forest carbon sinks within a country considered anthropogenic, or just a portion that is managed?

https://eciu.net/netzerotracker

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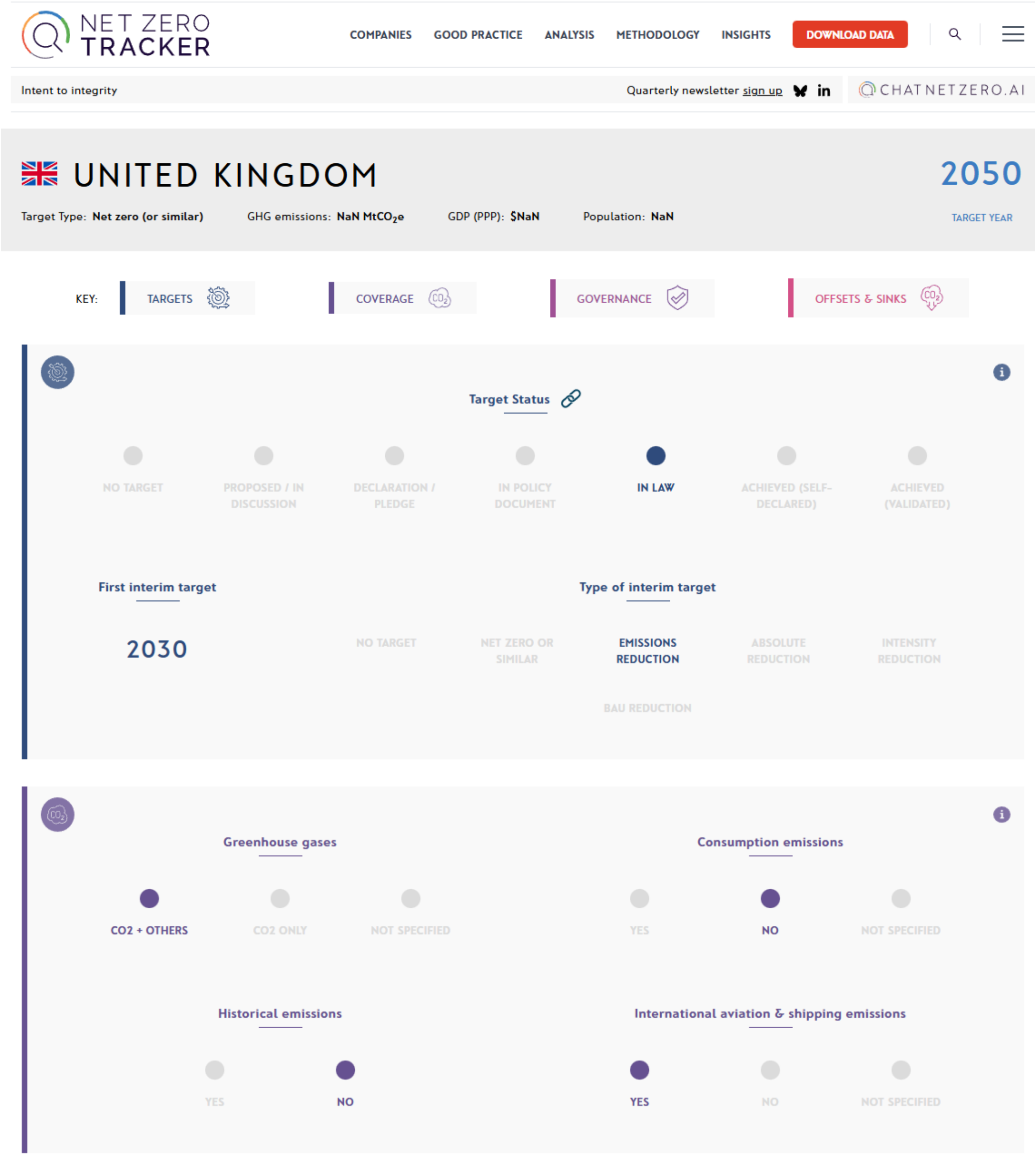
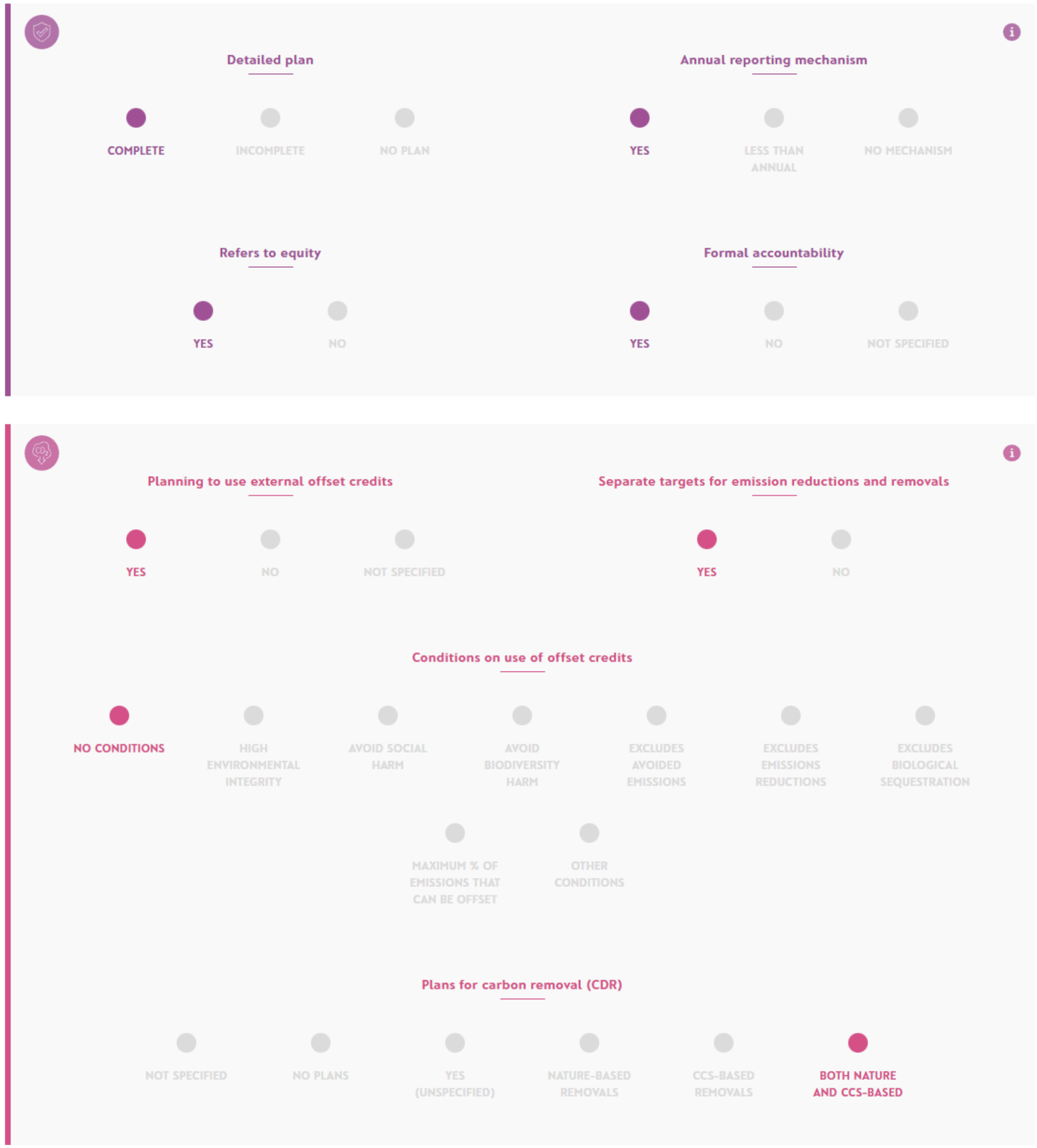
Net zero for countries





https://eciu.net/netzerotracker

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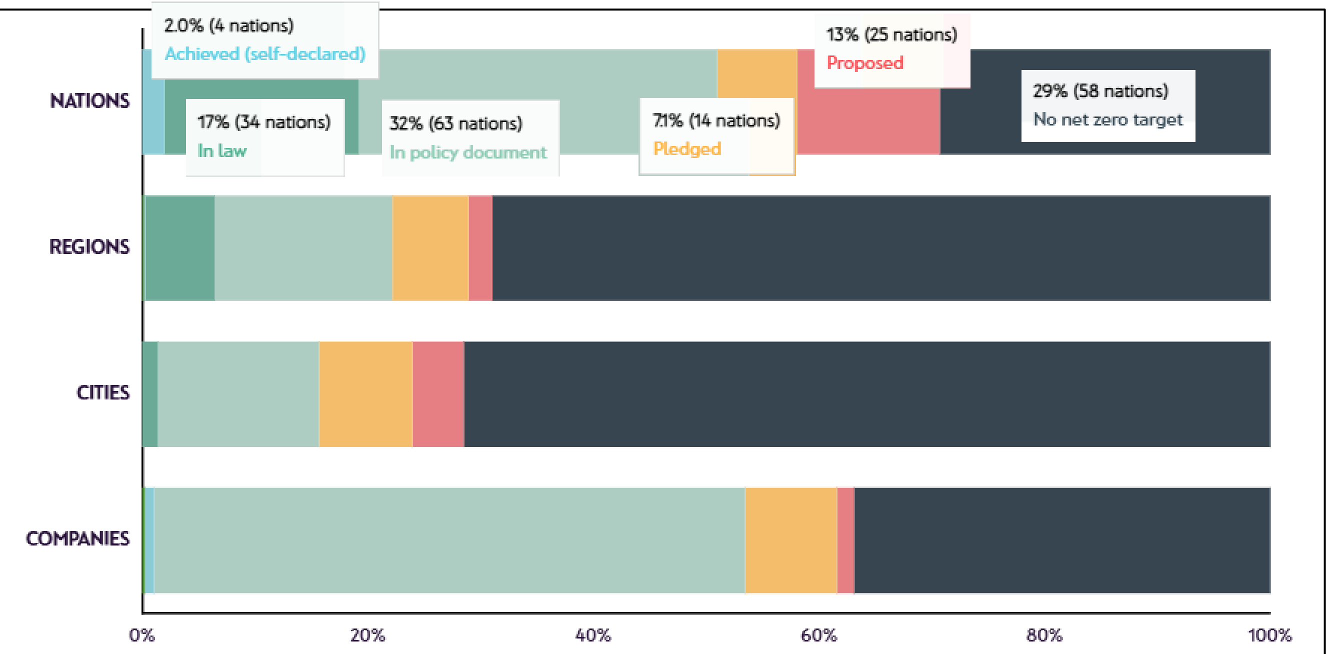
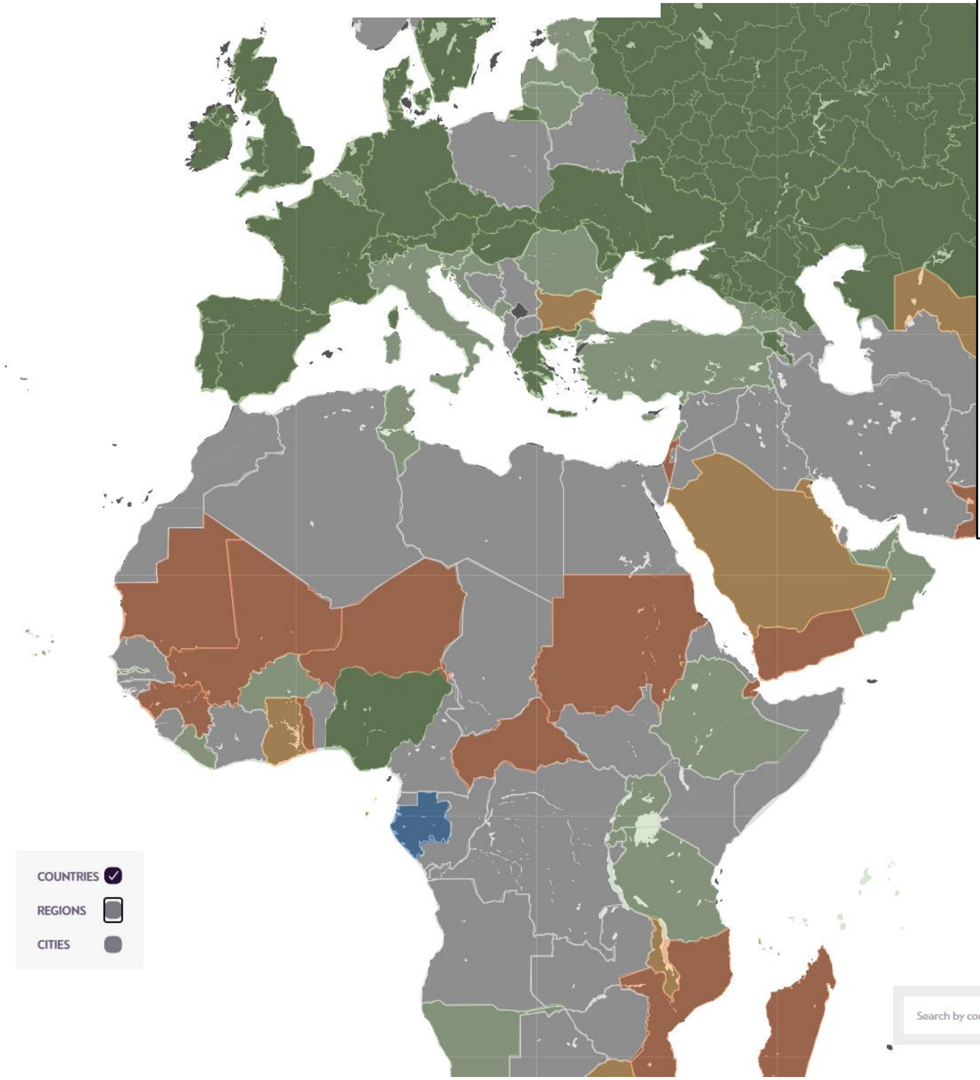
NET ZERO TRACKER



We analyse all nations that are parties to the UNFCCC, every region in the largest 25 emitting nations, all cities with more than 500,000 inhabitants, and the world's largest 2,000 publicly listed companies by annual revenue.



Imperial College London

https://zerotracker.net/

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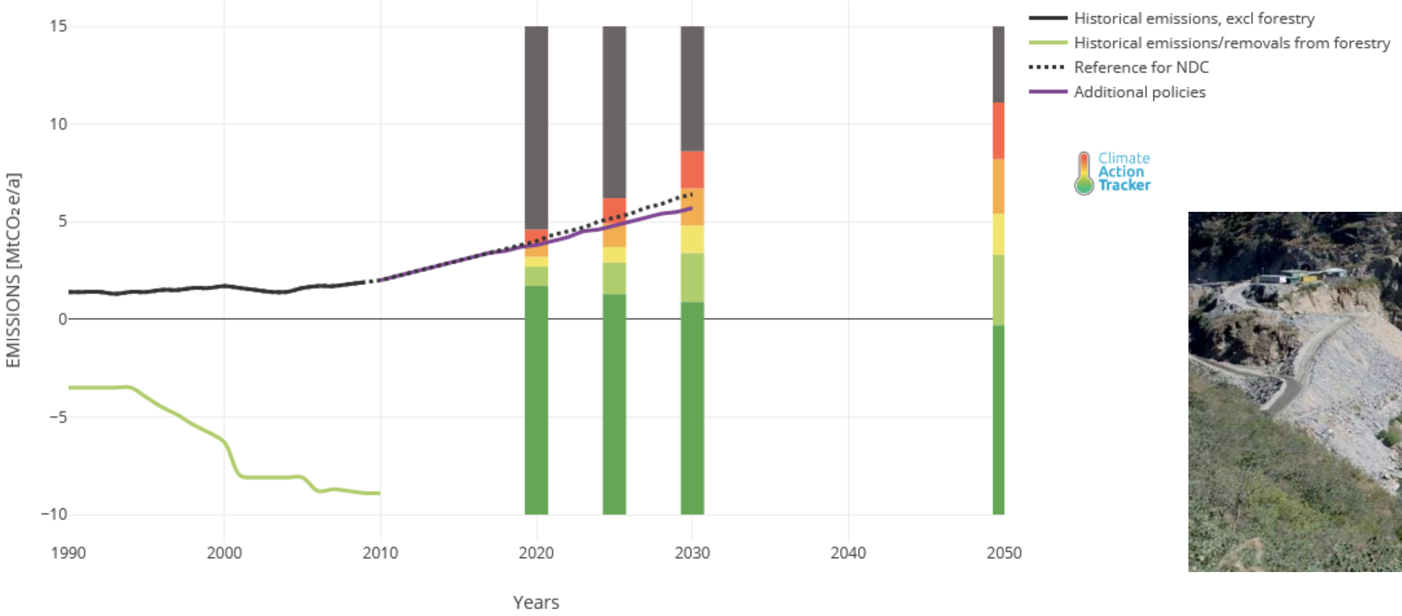
Bhutan



https://climateactiontracker.org/countries/bhutan/
https://www.researchgate.net/publication/329261835_Carbon_neutral_policy_in_action_the_case_of_Bhutan

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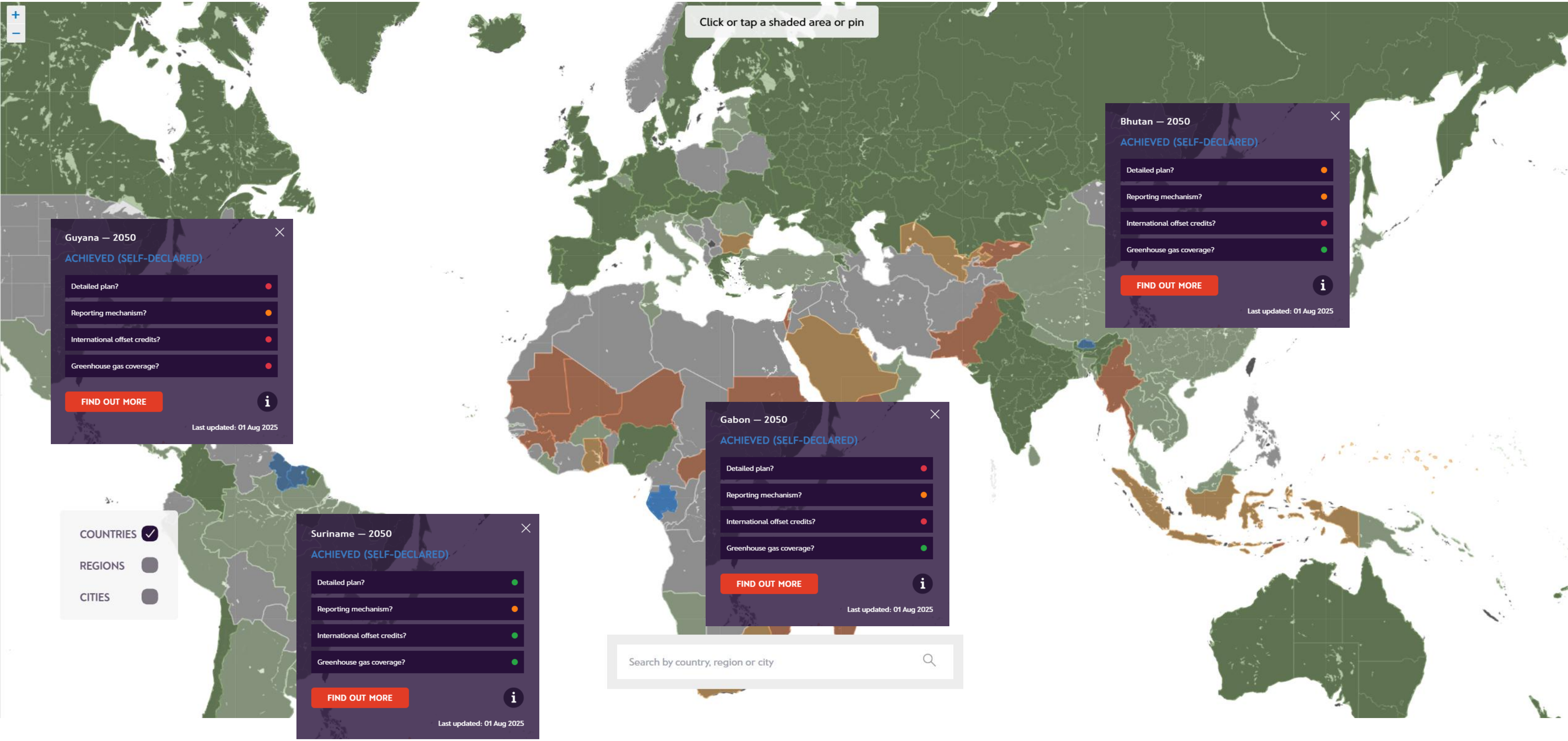

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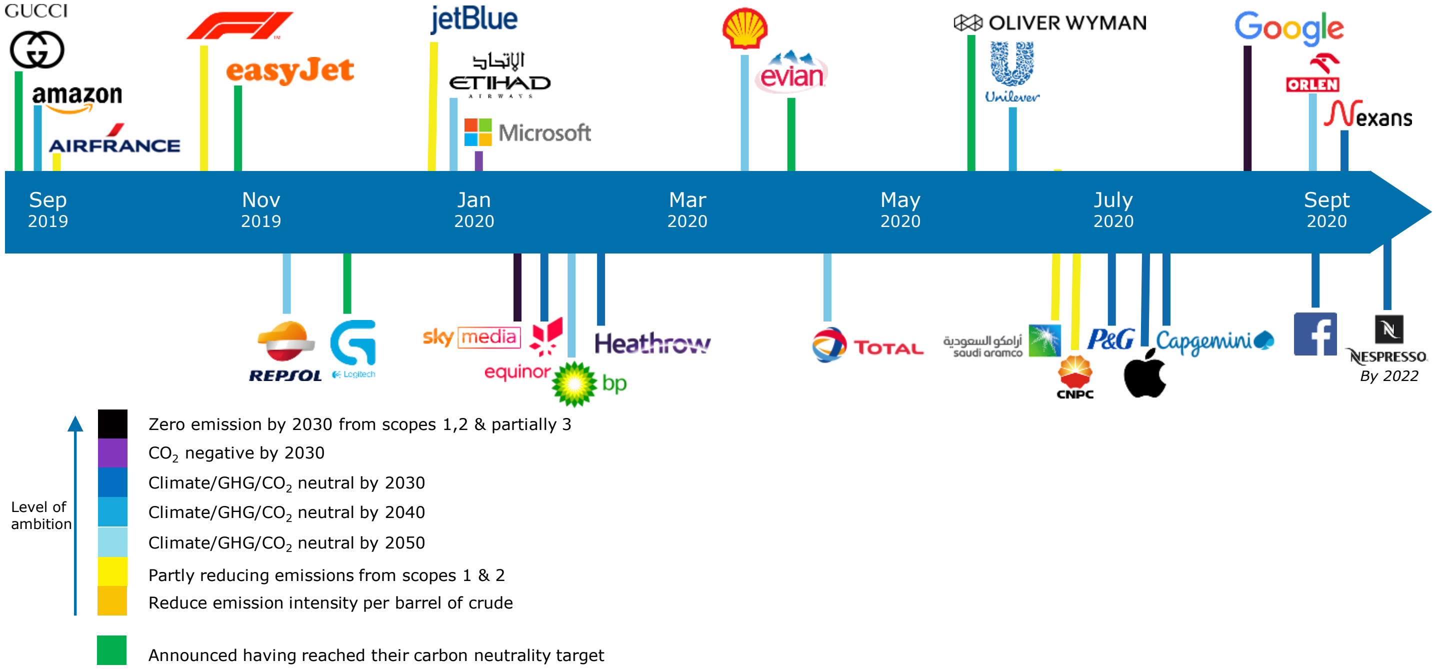
Net zero for companies (Sept 2019 to Sept 2020)

THE CORPORATE WORLD ENGAGED …



But definitions were not always consistent

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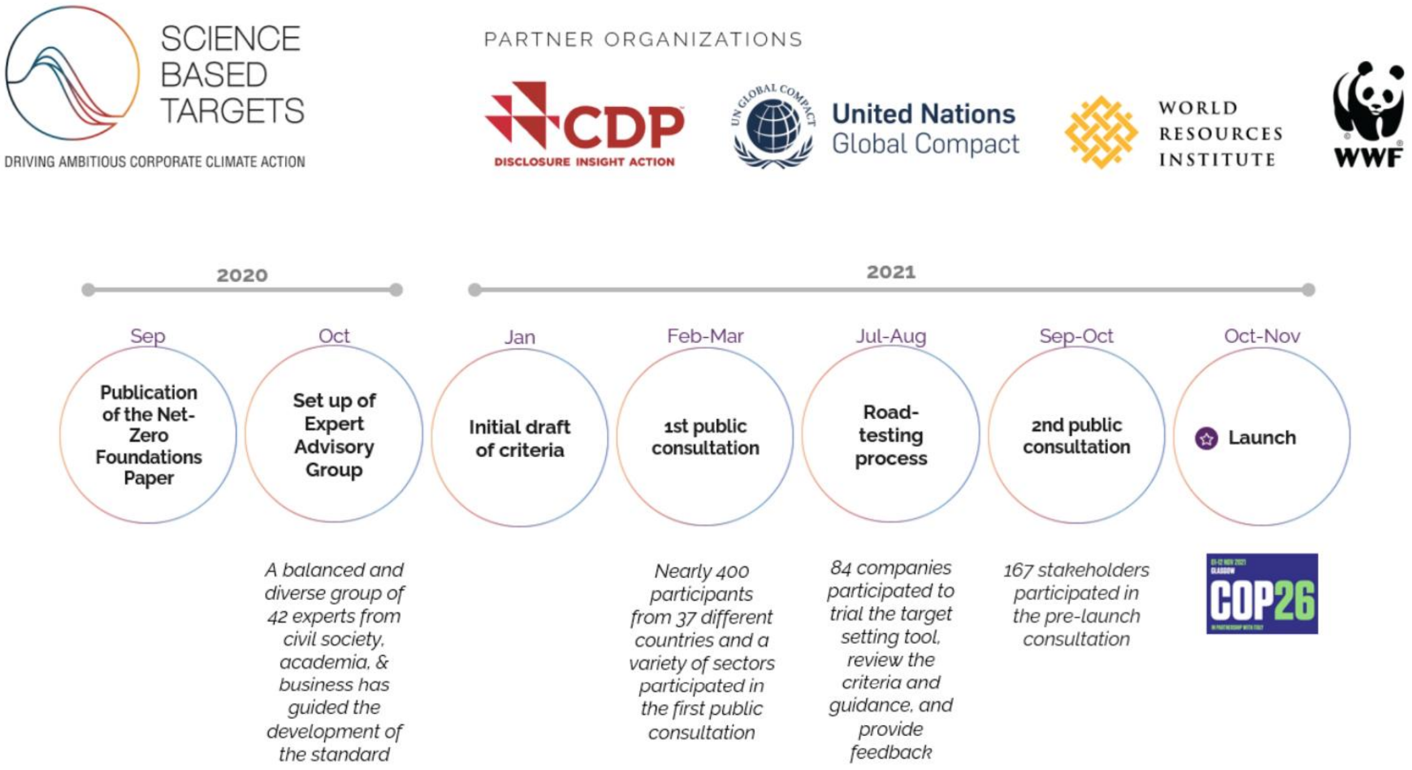


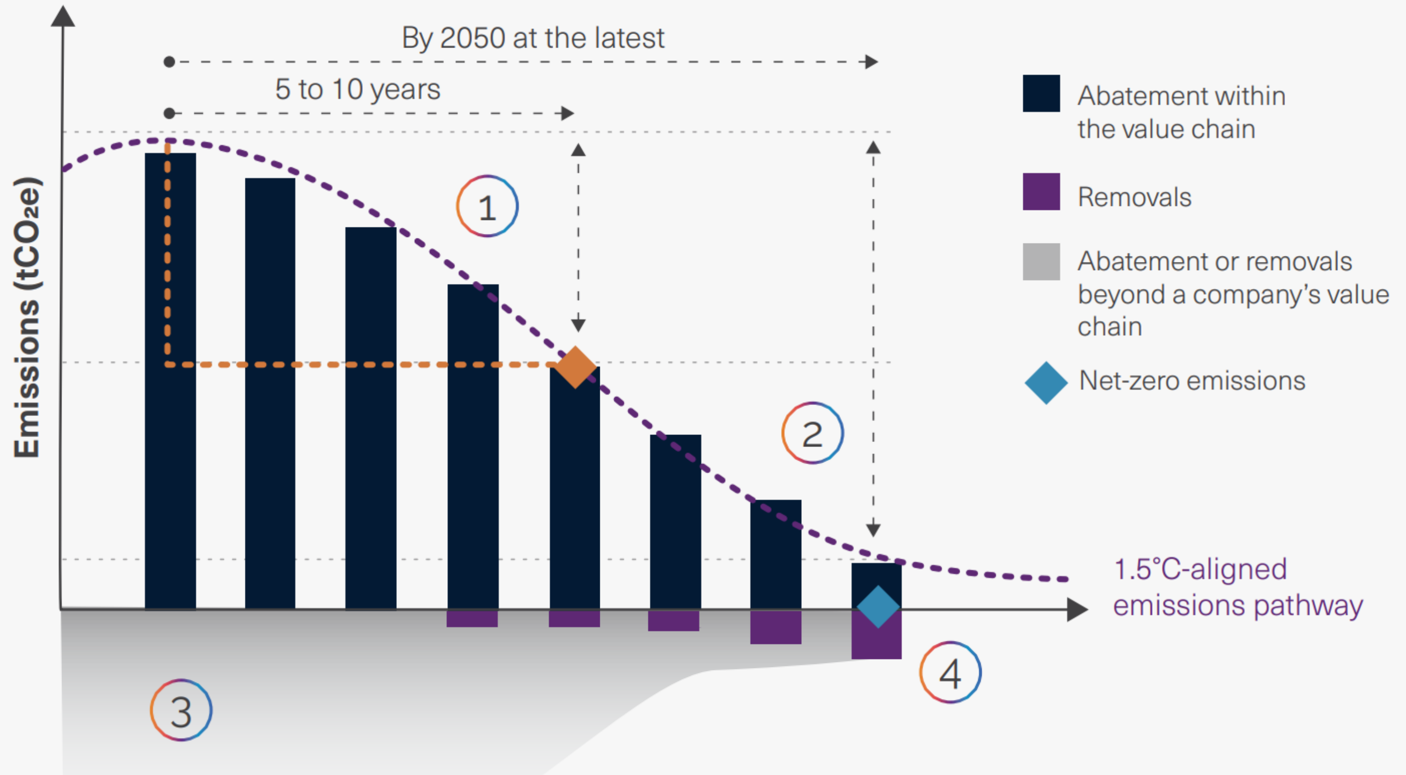
Figure 1 An outline of key milestones in Net-Zero Standard development process.

To contribute to societal net-zero goals, companies must deeply reduce emissions and counterbalance the impact of any emissions that remain. The SBTi Net-Zero Standard defines corporate net-zero as:

- Reducing scope 1, 2, and 3 emissions to zero or to a residual level that is consistent with reaching net-zero emissions at the global or sector level in eligible 1.5°C-aligned pathways
- Neutralizing any residual emissions at the net-zero target year and any GHG emissions released into the atmosphere thereafter.

https://sciencebasedtargets.org/resources/files/Net-Zero-Standard.pdf

Figure 2 Key elements of the Net-Zero Standard



1. **To set near-term SBTs:** 5–10 year emission reduction targets in line with 1.5°C pathways
2. **To set long-term SBTs:** Target to reduce emissions to a residual level in line with 1.5°C scenarios by no later than 2050
3. **Beyond value chain mitigation:** In the transition to net-zero, companies should take action to mitigate emissions beyond their value chains. For example, purchasing high-quality, jurisdictional REDD+ credits or investing in direct air capture (DAC) and geologic storage
4. **Neutralization of residual emissions:** GHGs released into the atmosphere when the company has achieved their long-term SBT must be counterbalanced through the permanent removal and storage of carbon from the atmosphere.

https://sciencebasedtargets.org/resources/files/Net-Zero-Standard.pdf

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Sectors

Currently, companies in all sectors apart from fossil fuels can set science-based targets using our cross-sector methods. However, we also develop sector-specific guidance for companies in some heavy emitting industries. Tailored to the needs and context of each industry, these enable companies to develop ambitious and achievable science-based targets aligned with 1.5°C.

If your sector is not listed or the sector-specific guidance is not yet finalized, you should use our core methodologies and resources to set your targets. See our Getting Started Guide to learn which methodology your company should use, according to your sector and other criteria.

View our Sector Guidance Summary for an overview of the pathway and guidance resources that are developed or upcoming for each sector.

Select your sector to view specific requirements and guidance.

Sector	Status	
Aluminum	Finalized	VIEW MORE
Apparel and footwear	In Development	VIEW MORE
Aviation	Finalized	VIEW MORE
Buildings	Finalized	VIEW MORE
Chemicals	In Development	VIEW MORE
Cement	Finalized	VIEW MORE
Financial institutions	Finalized	VIEW MORE
Forest, Land and Agriculture (FLAG)	Finalized	VIEW MORE
Information and Communication Technology (ICT)	Finalized	VIEW MORE
Land Transport	Finalized	VIEW MORE
Maritime	Finalized	VIEW MORE
Oil and Gas	In Development	VIEW MORE
Power	In Development	VIEW MORE
Steel	Finalized	VIEW MORE

https://sciencebasedtargets.org/standards-and-guidance#sectors

Carbon neutral (one view) vs net zero

The terms carbon neutrality and net zero emissions often reflect the same intention, i.e. neutralising the impact of human activity on the climate system. Below is a comparison against the key criteria based on current best practice.

	Carbon Neutral	**Net Zero (reduce first … then)**
Science	All human CO_2 Emissions released are balanced by CO_2 removal	All human GHG Emissions released are balanced by GHG removal
Scope	Direct Operations Purchased Energy Value Chain – selective*	Direct Operations Purchased Energy Value Chain – whole value chain with at least 66% of emissions covered by science-based targets
Reduction Targets	Set credible emissions reduction targets** and demonstration of progress through a carbon plan	Set and reduce actual emissions at a rate consistent with an SBTi 1.5 degree pathway
Offsetting Credits***	Credible and certified avoidance and/or removal projects i.e. cook stoves, water purification and re-forestation	Credible and certified carbon removal projects only – i.e. re- forest, land restoration, carbon capture technology
Time frame	Short Term	Long Term

*following CarbonNeutral Protocol 2020 Standard: https://www.carbonneutral.com/the-carbonneutral-protocol/5-steps-to-achieving-carbonneutral-certification/step-1-define/requirements
** preferably Science Based Targets *** Alternative solutions to credit are viable such as removal in you value chain i.e. invest in a forest on your or suppliers land

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Summary - A few essentials for net zero commitments

1. **A 1.5^{o}C trajectory** - the company must be reducing emissions from it's wider value chain in line with a 1.5^{o}C approved science-based target

2. **Neutralisation is allowed (at the end)** - High quality carbon removal are accepted as a means of balancing emissions but net zero only really achieved once emissions reduced to final "residual"

3. **Long term strategy (but not too long term)** - A net zero commitment should involve a long term strategy that is clearly and consistently communicated by the company, but should not be beyond 2050.

4. **The business model** of the company should still be viable in a net zero economy (addressing a company's transitions risks)

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Not without controversy

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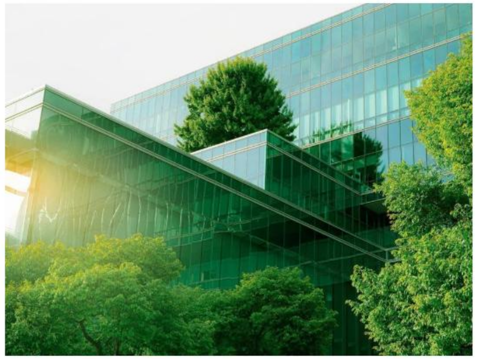
Article • Net Zero

Why SBTi Has Delisted More Than 200 High Profile Companies

By James Darley

January 17, 2025 • 6 mins



The Science-Based Targets initiative (SBTi) has officially removed over 200 high-profile companies from its list after they failed to meet the targets they set for emissions reductions | Credit: SBTi

In 2024 SBTi removed the commitments of 239 major global companies including Unilever, Walmart & Microsoft for failing to set near-term or net zero targets

The Science Based Targets initiative (SBTi), one of the world's foremost frameworks for corporate decarbonisation, removed more than 200 companies from its list of net zero commitments in 2024.

The decision, announced as part of the final report for the Business Ambition for 1.5°C campaign, spoke to the challenges faced by corporations in meeting stringent climate action requirements set out by international governing bodies, as well as self-imposed interim emissions targets.

Amongst the delisted companies in question are major players such as Microsoft, Unilever, Procter & Gamble and Walmart, many of which were prominent supporters of climate initiatives.

The SBTi's decision deals reputational blows to each of these companies, tarnishing the sustainability of their brands, though the SBTi has kept the door open for them to resubmit through the initiative in the future.

https://sustainabilitymag.com/articles/why-sbti-has-delisted-more-than-200-high-profile-companies

Hundreds of businesses drop net-zero commitments due to Scope 3 challenges

Almost 240 businesses did not follow through on their commitment to set verified net-zero targets or 1.5C-aligned science-based emissions targets, with many citing Scope 3 emissions and uncertainties about the future as major challenges.

Since 2019, more than 1,000 companies joined the Business Ambition for 1.5C campaign, an effort to raise the ambition on climate action and push companies to set science-based targets aligned with 1.5C as opposed to 2C or well-below 2C.

Out of these companies, 971 were analysed for a recent report assessing the campaign's impact. The study, from the Science-Based Targets Initiative (SBTi) excluded companies that underwent mergers, acquisitions, or awaited updated sector-specific guidance.

According to the findings, 84% of the analysed companies either set a target or are in the validation process.

Additionally, 96% of the surveyed companies rated the value of having science-based targets as good to very good, and 71% strongly agreed or agreed that the campaign delivered on their motivations for joining.

The top five countries for companies actively participating in the campaign are the UK, the US, France, Sweden and Germany.

https://www.edie.net/hundreds-of-businesses-drop-net-zero-commitments-due-to-scope-3-challenges/

Not without controversy

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POLICY & ECONOMICS

The SBTi Is No Longer Fit For Purpose

Opinion Article

BY CHRIS HOCKNELL | APR 15TH 2025 | 5 MINS

EARTH.ORG IS POWERED BY OVER 150 CONTRIBUTING WRITERS

"The slinking exodus from SBTi's framework is not necessarily a failure of corporate ambition. Rather, it is an indication that the existing model is too rigid to function in the real world, and always was," writes Chris Hocknell.

—

The Science-Based Targets initiative (SBTi) was once the gold standard for corporate climate action. It provided businesses with a structured framework to align their decarbonization strategies with the Paris Agreement, promising an approach to net zero that was both science-based and credible. However, rather predictably, cracks have started to appear.

Increasingly, companies are quietly withdrawing from their SBTi commitments, raising a fundamental question: is the net-zero model still fit for purpose?

The idea of "corporate net zero" has always been aspirational. The reality is that most business models rely on increasing growth – increasing production, or sales, or both necessitates increased input.

Whether that is literal energy in manufacturing, or other forms of increasing energy usage – through expanding offices, for example – growth will always be almost impossible to align with the prevailing ideal of "net zero".

Recent data supports this. For many businesses, net zero is not just hard; it is proving to be practically unachievable.

A 2024 analysis of 1,883 international firms found that 36% of them were already failing to meet their Scope 1 and 2 reduction targets, while 51% were struggling with Scope 3 emissions. Similarly, a 2024 report by Funds Europe's Carbon Impact Research revealed that nearly half of asset managers have failed to disclose their SBTi allocations.

So, why are companies abandoning their commitments?

The answer lies in a mix of political pressures, economic realities, and the diminishing returns of emissions reduction efforts.

Of course, the political mood that has swept across the world in the wake of Donald Trump's election has hardly helped to drive the corporate decarbonization agenda. Trade wars, high inflation and high interest rates have made further investments in next-generation clean technologies far less feasible.

Yet the slowdown also reflects far more fundamental hurdles for companies. SBTi's requirement for a 90% absolute reduction across Scope 1, 2, and 3 emissions is proving unworkable.

In reality, businesses have already implemented the low-hanging fruit. Cost-effective and easy-to-implement measures such as energy efficiency improvements and renewable energy sourcing make economic sense for businesses to adopt: they could end up saving them money, as well as saving their carbon budget.

Yet, the deeper the carbon cuts get, the more expensive they are, and the returns they deliver are ever more diminishing.

Of course, some businesses remain wary of regulatory flip-flopping, and prefer to "green-hush": continuing their internal sustainability efforts but avoiding public commitments. Staying silent makes these companies indistinguishable from those that simply drop their commitments. Meanwhile, the global economic headwinds of high interest rates and capital constraints are making large-scale investment in clean technology less viable.

This tension is evident in the energy sector, where rising fossil fuel prices and geopolitical instability are forcing companies to rethink their commitments.

BP, for example, recently scrapped its target to cut fossil fuel output by 40%, choosing instead to reinvest in oil and gas. Similarly, Shell abandoned its Scope 3 emissions reduction target under shareholder pressure, admitting that profitability takes precedence over climate pledges.

https://earth.org/the-sbti-is-no-longer-fit-for-purpose/

https://26227852.fs1.hubspotusercontent-eu1.net/hubfs/26227852/Why%20Companies%20are%20Dropping%20Net-Zero.pdf

Salesforce faces incredulity over “net zero” claims

The Stack

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SALESFORCE — EMISSIONS — NEWS

Salesforce faces incredulity over “net zero” claims

"We believe that companies should be able to claim they’re net zero if they’re fully committed to decarbonization..."

ED TARGETT
September 21, 2023 . 4:56 PM — 3 min read

Salesforce faced incredulity this week for conjuring up its own definition of achieving “net zero” – the $207 billion software business saying that

“Salesforce believes that — as a planet — we must reduce our emissions by 90% and do so as quickly as possible. But, realistically, we may not achieve this milestone until 2040-50... Under SBTi’s definition, no company can claim they are net zero for decades. We believe that companies should be able to claim they’re net zero if they’re fully committed to decarbonization and have demonstrated they’re on the path to achieving the deep emissions reductions that will put the planet on a 1.5 degree trajectory, if they’re also compensating for their remaining emissions with high-quality carbon credits to achieve net zero residual emissions...”

That approach was ridiculed this week by one experienced GreenOps specialist, Mark Butcher, who said on LinkedIn: “Personally I don't think companies should be able to simply invent a new piece of terminology just because they don’t like the version used by everyone else as it doesn't suit their narrative... Using their logic, can you all please congratulate me on winning the 100m gold in the 2036 Olympics. Obviously according to Salesforce this is OK because I started running last week, and I'm definitely going to meet the required standard.

“This kind of marketing led nonsense really undermines any good progress they might actually be making and makes it impossible for their customers to accurately report their own emissions.”

https://www.thestack.technology/salesforce-net-zero-claims-sbti/

Net zero: discussion points

- *"Forget about net zero, we need real zero"* said Greta Thunberg. Do you agree?

- Is a net zero commitment taking a leadership position?

Discuss in your groups and select one person to report back…

https://www.bbc.co.uk/news/av/world-51193460/davos-forget-about-net-zero-we-need-real-zero-greta-thunberg

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Carbon Accounting

The Challenge of Comparability

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The challenge of comparability

- Absence of one international accounting standard – or even one defined approach within the standard
- Even within a company with one chosen framework, there is a balance between flexibility and comparability
- Reliance on estimation and differences in estimation methods
- High level of uncertainty – +/-20% typical even for direct sources
- Lack of completeness and/or transparency in methodology
- Differences in company structure and approach

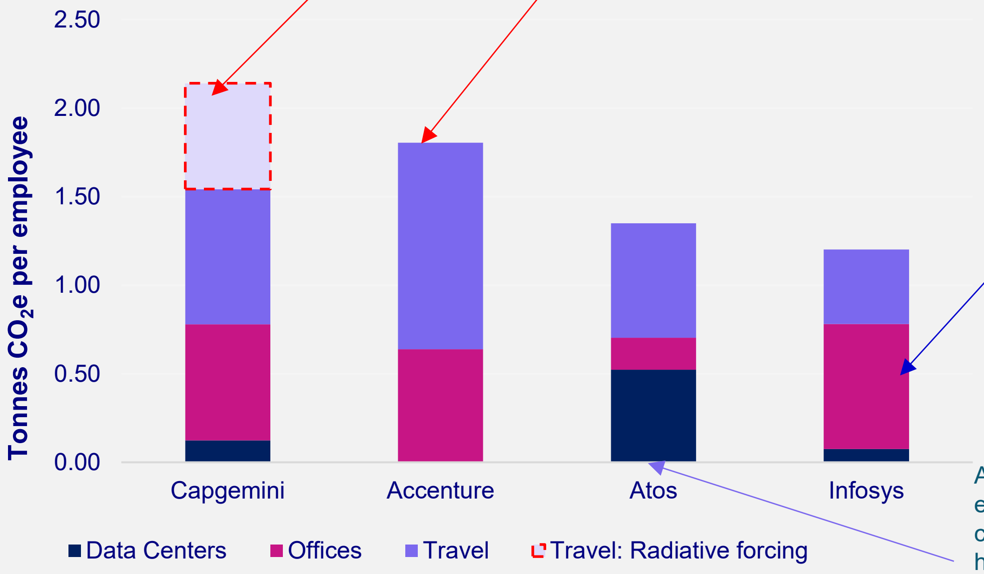
Additional reading: *Unpacking carbon accounting numbers: A study of the commensurability and comparability of corporate greenhouse gas emission disclosures*

- Matthew Wegener a, Real Labelle b, Lambert Jerman

Example of some of the comparability issues: breakdown of emissions per employee in 2018, Capgemini & three peers

Reporting on impact of air travel radiative forcing is recommended best practice, but Capgemini appears to be only company following it.

Taking into consideration the differences in calculation approach, it's likely Accenture have the highest impact per employee from business travel.



Infosys have the highest impact from offices, likely due to high presence in India and due to inclusion of all leased offices (Capgemini take same approach of including all leased offices regardless of control level, Accenture & Atos do not disclose their approach on this topic).

Atos have the largest data center emissions as they provide more data center services that the other 3 but they have relatively low emissions from offices – this is most likely due to high proportion of operations in France, and low proportion in India.

The emissions included have been made as comparable as possible, there are likely to be significant differences in calculation approach. Infosys do not directly disclose emissions from data centres to CDP so a breakdown has been assumed here based on info in their annual report. All electricity emissions reported here are based on a location-based (grid average) approach, to help with comparability.

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